PET and Gestational Hypertension 04: Prediction and Prevention

Companion reading handout for simultaneous listening.
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Opening Case

Let us begin before the patient is hypertensive.

She is twelve weeks. Her blood pressure is normal. She has chronic hypertension in her chart, a prior pregnancy complicated by early pre-eclampsia, and a body mass index in the obese range. Nothing is happening today. But the future pregnancy has already declared a risk pattern.

Prevention in pre-eclampsia is not about guessing perfectly. It is about identifying patients whose placental and vascular risk is high enough that low-risk interventions are worth starting early.

Clinical Risk Screening

The first layer is clinical risk.

High-risk factors across major guidelines include prior pre-eclampsia, chronic hypertension, chronic kidney disease, pregestational diabetes, autoimmune disease such as SLE or antiphospholipid syndrome, and multifetal gestation. Moderate-risk factors include nulliparity, obesity, maternal age over forty, family history, a long interpregnancy interval, assisted reproduction, and elevated early pregnancy blood pressure.

Clinical risk factors are useful because they are available everywhere. But they are not highly sensitive for preterm pre-eclampsia. ISSHP notes that multivariable screening performs better when it combines maternal factors, blood pressure, uterine artery Doppler, and PlGF. The Israeli position paper also discusses first-trimester algorithms that detect much of early pre-eclampsia, but with low positive predictive value in low-risk populations and dependence on local resources.

So the first teaching point is this: risk-factor screening is the minimum. Multivariable screening is more biologically rich where it is available and quality controlled.

Positive predictive value is not a statistical decoration here. It changes patient care. A screening model can detect many cases and still produce many false positives when the outcome is uncommon. That means counseling, aspirin decisions, follow-up intensity, and anxiety must be handled carefully. The goal is not to label a normal twelve-week patient as diseased. The goal is to identify enough excess risk to justify a safe preventive intervention and a smarter surveillance plan.

Aspirin

Now let us talk about aspirin.

Aspirin is the main pharmacologic prevention tool. But it works best when you understand what problem it is trying to modify.

In pre-eclampsia, there is abnormal placentation, platelet activation, vasoconstriction, inflammation, and endothelial dysfunction. Aspirin irreversibly inhibits cyclooxygenase one. That lowers thromboxane A two, a platelet-derived vasoconstrictor and pro-aggregatory signal. It helps rebalance the thromboxane to prostacyclin relationship toward less platelet aggregation and less vasoconstriction.

SOMANZ adds another layer. Aspirin may also trigger anti-inflammatory lipoxins, increase interleukin ten and nitric oxide, and reduce tumor necrosis factor alpha. So aspirin is not just "a blood thinner." It is an antiplatelet and anti-inflammatory intervention aimed at placental vascular development and maternal endothelial stress.

Timing matters.

The benefit is strongest when aspirin starts before sixteen weeks. That makes physiologic sense. Spiral artery remodeling and early placental development are still being shaped. If you begin after the disease architecture is already established, the prevention window is narrower.

SOMANZ reviewed randomized trials and found that aspirin started before sixteen weeks reduced pre-eclampsia, early-onset pre-eclampsia, preterm delivery, and small-for-gestational-age infants. Starting at or after sixteen weeks showed less benefit.

Dose matters too, but guidelines differ.

ACOG commonly recommends eighty one milligrams daily for high-risk patients, started between twelve and twenty eight weeks, optimally before sixteen weeks. NICE recommends seventy five to one hundred fifty milligrams from twelve weeks for patients with risk factors. ISSHP recommends one hundred fifty milligrams at night after multivariable screening, or one hundred to one hundred sixty two milligrams per day after clinical-risk and BP screening. SOMANZ strongly recommends one hundred fifty milligrams daily for high-risk women, preferably before sixteen weeks, often at bedtime.

Why have newer guidelines moved toward higher doses?

Pregnancy increases platelet turnover. Higher body mass may reduce aspirin effect. Some patients have apparent platelet insensitivity at lower doses. The ASPRE-style evidence supports one hundred fifty milligrams nightly for screen-positive women, especially for preventing preterm pre-eclampsia.

Night dosing is not just a ritual in these protocols. It comes from the idea that platelet activation and blood pressure biology may have circadian behavior, and from the way major prevention trials were designed. That does not mean every patient everywhere must receive one hundred fifty milligrams at night, but it explains why several international protocols prefer that strategy for high-risk or screen-positive patients.

But do not turn this into universal aspirin without thought. Universal aspirin for low-risk nulliparas is not routine in the guidelines because benefit is small, adherence may be lower, and bleeding signals exist. Prevention is still risk-based.

Stopping aspirin also varies. ISSHP suggests stopping by thirty six weeks. SOMANZ recommends individualized cessation between thirty four weeks and delivery, considering bleeding risk and shared decision-making. Local practice matters.

Calcium

Now calcium.

Calcium is not aspirin. Its mechanism is less certain, and its benefit depends strongly on baseline intake.

The working model is that low calcium intake may stimulate parathyroid hormone or renin release. That can increase vascular smooth muscle intracellular calcium and promote vasoconstriction. Supplementation may reduce this vasoconstrictive tendency and improve uteroplacental resistance.

ISSHP recommends at least five hundred milligrams per day for women whose dietary calcium intake is below nine hundred milligrams per day. SOMANZ recommends supplementation when dietary intake is below one gram per day and reports that high-dose supplementation above one gram per day reduces pre-eclampsia and gestational hypertension in low-intake women. In adequate-intake populations, benefit is not established.

So the resident memory point is simple: calcium prevents pre-eclampsia when calcium intake is low. It is not a universal vascular drug for everyone.

Exercise

Exercise is another prevention tool, but again, context matters.

ISSHP cites randomized trial evidence that exercise reduces gestational hypertension and pre-eclampsia. SOMANZ recommends two and a half to five hours per week of moderate-intensity aerobic, stretching, or resistance exercise as routine pregnancy wellbeing with hypertensive-disorder benefit.

But established pre-eclampsia is different. Once the disease exists, exercise is contraindicated, and gestational hypertension requires caution depending on severity and local guidance. Exercise is prevention and general health, not treatment of active disease.

What Not to Use

Now let us talk about what not to use.

This is where mechanisms can mislead you.

Salt restriction sounds plausible. Oxidative stress makes vitamins C and E sound plausible. Angiogenic imbalance makes statins and proton pump inhibitors sound plausible. Coagulation and placental thrombosis make LMWH sound plausible. Inflammation makes fish oil and other supplements sound plausible.

But plausible is not proven.

The corpus does not support routine use of salt restriction, fish oil specifically for pre-eclampsia prevention, garlic, vitamins C and E, oral magnesium, progesterone, statins outside research or another indication, LMWH in patients without antiphospholipid syndrome or another anticoagulation indication, proton pump inhibitors, clopidogrel, or similar novel agents solely to prevent pre-eclampsia.

This is an important professional habit. Do not teach mechanisms as if they are outcomes. A mechanism is a reason to study an intervention. It is not a reason to prescribe it when trials do not show patient benefit.

Closing Synthesis

Let us close with the prevention hierarchy.

First, identify risk early. At minimum, use clinical risk factors. Where available and validated, consider multivariable first-trimester screening.

Second, use aspirin for patients at increased risk, started early, ideally before sixteen weeks, with dose and stopping time aligned to local guideline.

Third, address calcium only when intake is low.

Fourth, encourage pregnancy-appropriate exercise before disease develops.

Fifth, do not add unproven therapies simply because the mechanism sounds attractive.

Resident Reflex

The resident reflex is this: prevention is won early. Aspirin belongs before the disease declares itself. Calcium belongs where intake is low. Exercise belongs before active disease. And plausible biology must still answer to outcome evidence.